

Solution Architectur

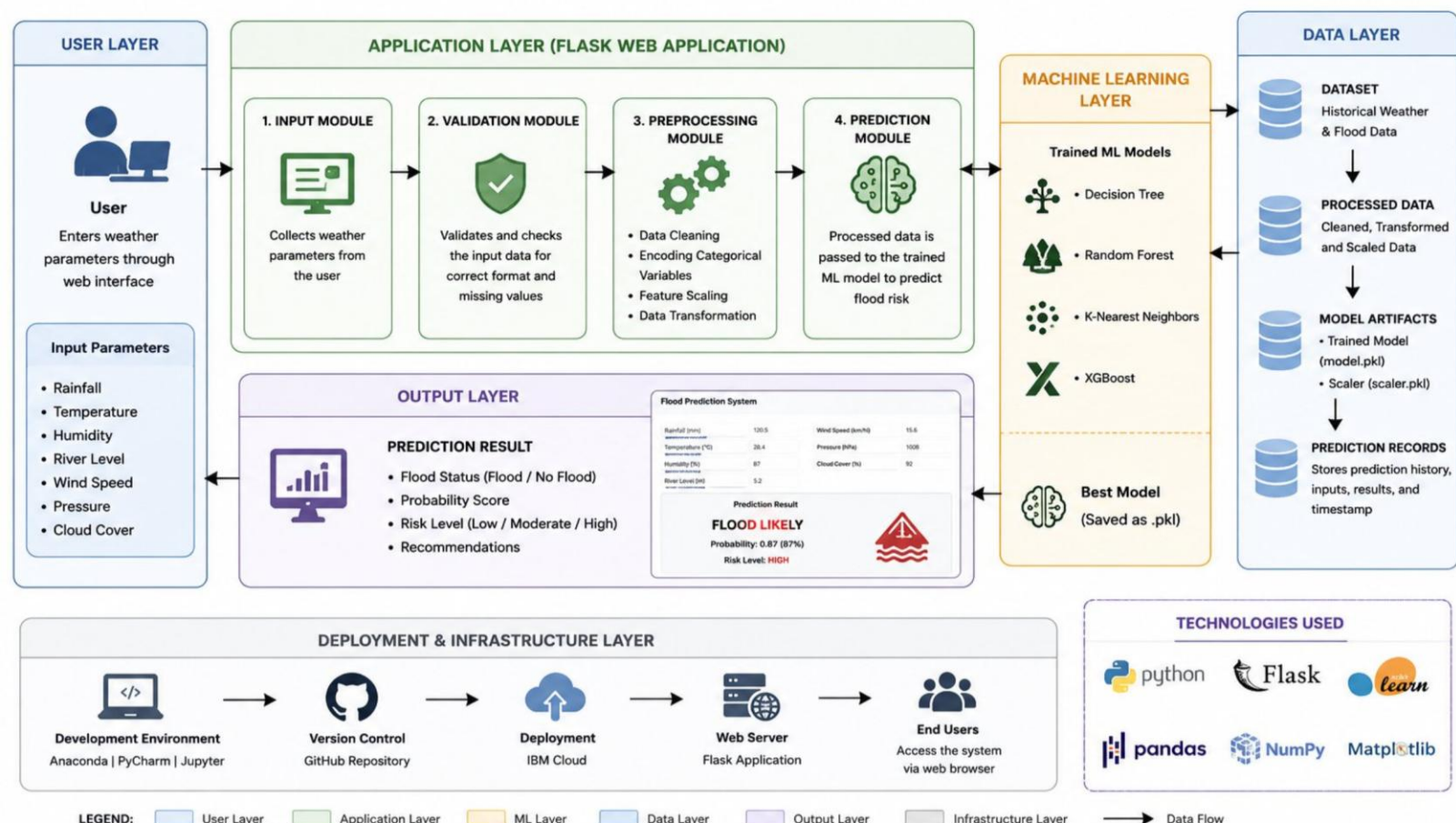
Date	4 July 2026
Team ID	X X X X
Project Name	Raising Waters
Maximum Mark	5 Mark s

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the st layout below as a template to map out your Presentation Layer, Application Layer, and D Layer. Fill in the specific technologies and components for your project directly into create your arc hitectural diagram

FLOOD PREDICTION SYSTEM ARCHITECTURE

Rising Waters: A Machine Learning Approach to Flood Prediction



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
User Interface	Provides an interactive and user-friendly interface	HTML, CSS, JavaScript
Application Layer	Handles user requests, processes input data, connects the web interface	Flask (Python)
Machine Learning Layer	Trains, evaluates, and deploys machine learning models	Scikit-learn, XGBoost, Joblib
Data Layer	Stores, manages, preprocesses, and analyzes historical	Pandas, NumPy, Flood Prediction Dataset